

# Лабораторная работа №10

Настройка SMTP-аутентификации, LMTP и TLS в Postfix + Dovecot

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## Цель работы

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Изучить расширенные механизмы конфигурирования почтового сервера, включая **LMTP**, **SMTP-аутентификацию** и **SMTP over TLS**, а также интегрировать Postfix и Dovecot в единое почтовое решение.

## Выполнение работы

---

```
22
23 # Protocols we want to be serving.
24
25 protocols = imap pop3 lmtp
26
27 # A comma separated list of IPs or hosts where to listen in for connections.
28 # "*" listens in all IPv4 interfaces, "::" listens in all IPv6 interfaces.
29 # If you want to specify non-default ports or anything more complex,
30 # edit conf.d/master.conf.
31 #listen = *, ::
32
33 # Base directory where to store runtime data.
34 #base_dir = /var/run/dovecot/
35
36 # Name of this instance. In multi-instance setup doveadm and other commands
37 # can use -i <instance_name> to select which instance is used (an alternative
38 # to -c <config_path>). The instance name is also added to Dovecot processes
39 # in ps output.
40 #instance_name = dovecot
41
```

Рис. 1: Добавление LMTP в список протоколов

```
56  
57 service lmtp {  
58     unix_listener /var/spool/postfix/private/dovecot-lmtp {  
59         group = postfix  
60         user = postfix  
61         mode = 0600  
62     }  
63 }  
64
```

Рис. 2: Настройка unix\_listener для LMTP

```
46
47 # Username formatting before it's looked up from databases. You can use
48 # the standard variables here, eg. %Lu would lowercase the username, %n would
49 # drop away the domain if it was given, or "%n-AT-%d" would change the '@' into
50 # "-AT-". This translation is done after auth_username_translation changes.
51
52 auth_username_format = %Ln|
53
```

Рис. 3: auth\_username\_format

```
Nov 29 08:31:41 server postfix/smtpd[21859]: connect from client.vabazlov.net[192.168.1.30]
Nov 29 08:31:41 server postfix/smtpd[21859]: 8611D46F93: client=client.vabazlov.net[192.168.1.30]
Nov 29 08:31:41 server postfix/cleanup[21863]: 8611D46F93: message-id=<20251129083141.7C29261320A5@client.vabazlov.net>
Nov 29 08:31:41 server postfix/smtpd[21859]: disconnect from client.vabazlov.net[192.168.1.30] ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Nov 29 08:31:41 server postfix/qmgr[21705]: 8611D46F93: from=<vabazlov@client.vabazlov.net>, size=541, nrcpt=1 (queue active)
Nov 29 08:31:41 server postfix/local[21864]: 8611D46F93: passing <vabazlov@vabazlov.net> to transport=lmtpl
Nov 29 08:31:41 server dovecot[21754]: lmtp(21866): Connect from local
Nov 29 08:31:41 server dovecot[21754]: lmtp(vabazlov)<21866><dpFnIW2vKmlqVQAAFnYJw>: msgid=<20251129083141.7C29261320A5@client.vabazlov.net>: saved mail to INBOX
Nov 29 08:31:41 server postfix/lmtp[21865]: 8611D46F93: to=<vabazlov@vabazlov.net>, relay=server.vabazlov.net[private:dovecot-lmtp], delay=0.02, delays=0/0.01/0, dsn=2.0.0, status=sent (250 2.0.0 <vabazlov@vabazlov.net> dpFnIW2vKmlqVQAAFnYJw Saved)
Nov 29 08:31:41 server dovecot[21754]: lmtp(21866): Disconnect from local: Logged out (state=READY)
Nov 29 08:31:41 server postfix/qmgr[21705]: 8611D46F93: removed
```

Рис. 4: Логи прохождения письма

```
[vabazlov@server.vabazlov.net ~]#  
[vabazlov@server.vabazlov.net ~]$ MAIL=~/.Maildir/ mail  
s-nail version v14.9.24.  Type '?' for help  
/home/vabazlov/.Maildir: 2 messages 1 unread  
      1 vabazlov                2025-11-29 08:13    18/632    "test1  
▶U  2 vabazlov@client.vaba  2025-11-29 08:31    21/815    "LMTP test  
&  
[-- Message  2 -- 21 lines, 815 bytes --]:  
Date: Sat, 29 Nov 2025 08:31:41 +0000  
To: vabazlov@vabazlov.net  
Subject: LMTP test  
Message-Id: <20251129083141.7C29261320A5@client.vabazlov.net>  
From: vabazlov@client.vabazlov.net  
  
.  
  
& q  
Held 2 messages in /home/vabazlov/.Maildir  
[vabazlov@server.vabazlov.net ~]$
```

Рис. 5: Просмотр полученных сообщений

```
83
84 service auth {
85     unix_listener auth-userdb {
86         mode = 0600
87         user = dovecot
88     }
89
90     unix_listener /var/spool/postfix/private/auth {
91         group = postfix
92         user = postfix
93         mode = 0660
94     }
95
96     # Auth process is run as this user.
97     #user = $default_internal_user
98 }
99
```

Рис. 6: service auth конфигурация

```
[root@server.vabazlov.net server]# cp /etc/dovecot/conf.d/10-master.conf
[root@server.vabazlov.net server]#
[root@server.vabazlov.net server]# postconf -e 'smtpd_sasl_type = dovecot'
[root@server.vabazlov.net server]# postconf -e 'smtpd_sasl_path = private/auth'
[root@server.vabazlov.net server]# postconf -e 'smtpd_recipient_restrictions = reject_unknown_recipient_domain, permit_mynetworks, reject_non_fqdn_recipient, reject_unauth_destination, reject_unverified_recipient, permit'
[root@server.vabazlov.net server]# postconf -e 'mynetworks = 127.0.0.0/8'
[root@server.vabazlov.net server]#
```

Рис. 7: postconf ограничения

# Параметры smtpd

```
1 #
2 # Postfix master process configuration file. For details on the format
3 # of the file, see the master(5) manual page (command: "man 5 master" or
4 # on-line: http://www.postfix.org/master.5.html).
5 #
6 # Do not forget to execute "postfix reload" after editing this file.
7 #
8 # =====
9 # service type private unpriv chroot wakeup maxproc command + args
10 # (yes) (yes) (no) (never) (100)
11 # =====
12 smtp inet n - n - - smtpd
13 #smtp inet n - n - 1 postscreen
14 #smtpd pass - - n - - smtpd
15 #dnsblog unix - - n - 0 dnsblog
16 #tlsproxy unix - - n - 0 tlsproxy
17 # Choose one: enable submission for loopback clients only, or for any client.
18 #127.0.0.1:submission inet n - n - - smtpd
19 #submission inet n - n - - smtpd
20 # -o syslog_name=postfix/submission
21 # -o smtpd_tls_security_level=encrypt
22 # -o smtpd_sasl_auth_enable=yes
23 # -o smtpd_tls_auth_only=yes
24 # -o local_header_rewrite_clients=static:all
25 # -o smtpd_reject_unlisted_recipient=no
26 # Instead of specifying complex smtpd_<xxx>_restrictions here,
27 # specify "smtpd_<xxx>_restrictions=$mua_<xxx>_restrictions"
28 # here, and specify mua_<xxx>_restrictions in main.cf (where
29 # "<xxx>" is "client", "helo", "sender", "relay", or "recipient").
30 # -o smtpd_client_restrictions=
31 # -o smtpd_helo_restrictions=
32 # -o smtpd_sender_restrictions=
33 # -o smtpd_relay_restrictions=
34 # -o smtpd_recipient_restrictions=reject_non_fqdn_recipient,reject_unknown_recipient_domain,permit_sasl_authenticated,reject
35 # -o milter_macro_daemon_name=ORIGINATING
36 # Choose one: enable submissions for loopback clients only, or for any client.
37 #127.0.0.1:submissions inet n - n - - smtpd
38 #submissions inet n - n - - smtpd
39 # -o syslog_name=postfix/submissions
40 # -o smtpd_tls_wrappermode=yes
41 # -o smtpd_sasl_auth_enable=yes
42 # -o local_header_rewrite_clients=static:all
```

## Проверка аутентификации

```
[root@client.vabazlov.net ~]# telnet server.vabazlov.net 25
Trying 192.168.1.1...
Connected to server.vabazlov.net.
Escape character is '^]'.
220 server.vabazlov.net ESMTP Postfix
EHLO test
250-server.vabazlov.net
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-STARTTLS
250-AUTH PLAIN
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250-SMTPUTF8
250 CHUNKING
AUTH PLAIN dmFiYXpsb3YAdmFiYXpsb3YAMTIzNDU2
235 2.7.0 Authentication successful
quit
221 2.0.0 Bye
Connection closed by foreign host.
[root@client.vabazlov.net ~]#
```

```
[root@server.vabazlov.net server]#  
[root@server.vabazlov.net server]# cp /etc/pki/dovecot/certs/dovecot.pem /etc/pki/tls/certs/  
[root@server.vabazlov.net server]# cp /etc/pki/dovecot/private/dovecot.pem /etc/pki/tls/private/  
[root@server.vabazlov.net server]#  
[root@server.vabazlov.net server]# postconf -e 'smtpd_tls_cert_file = /etc/pki/tls/certs/dovecot.pem'  
[root@server.vabazlov.net server]# postconf -e 'smtpd_tls_key_file = /etc/pki/tls/private/dovecot.pem'  
[root@server.vabazlov.net server]# postconf -e 'smtpd_tls_session_cache_database = btree:/var/lib/postfix/smtpd_scache'  
[root@server.vabazlov.net server]# postconf -e 'smtpd_tls_security_level = may'  
[root@server.vabazlov.net server]# postconf -e 'smtp_tls_security_level = may'  
[root@server.vabazlov.net server]# █
```

Рис. 10: TLS конфигурация и настройка Postfix

## Добавление службы submission (порт 587)

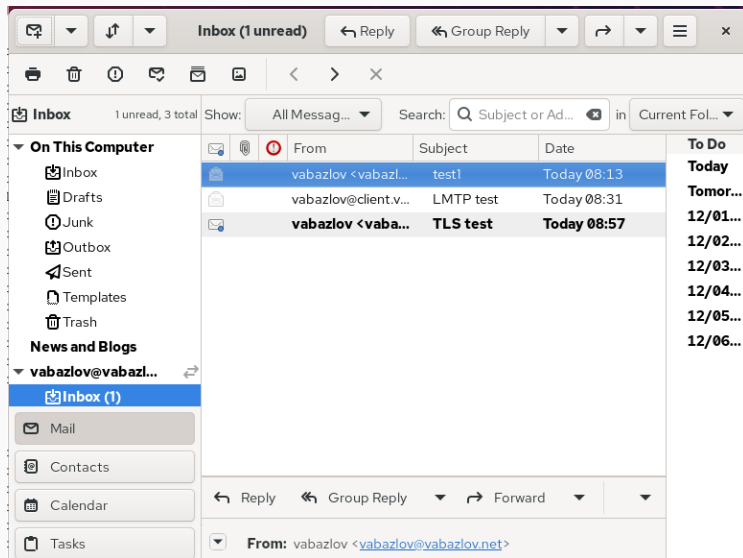
```
1 #
2 # Postfix master process configuration file. For details on the format
3 # of the file, see the master(5) manual page (command: "man 5 master" or
4 # on-line: http://www.postfix.org/master.5.html).
5 #
6 # Do not forget to execute "postfix reload" after editing this file.
7 #
8 # -----
9 # service type private unpriv chroot wakeup maxproc command + args
10 # (yes) (yes) (no) (never) (100)
11 # -----
12 smtp inet n - n - - smtpd
13 #smtp inet n - n - 1 postscreen
14 #smtpd pass - - n - - smtpd
15 #dnsblog unix - - n - 0 dnsblog
16 #tlsproxy unix - - n - 0 tlsproxy
17 # Choose one: enable submission for loopback clients only, or for any client.
18 #127.0.0.1:submission inet n - n - - smtpd
19 submission inet n - n - - smtpd
20 # -o syslog_name=postfix/submission
21 # -o smtpd_tls_security_level=encrypt
22 # -o smtpd_sasl_auth_enable=yes
23 # -o smtpd_tls_auth_only=yes
24 # -o local_header_rewrite_clients=static:all
25 # -o smtpd_reject_unlisted_recipient=no
26 # Instead of specifying complex smtpd_<xxx>_restrictions here,
27 # specify "smtpd_<xxx>_restrictions=$mua_<xxx>_restrictions"
28 # here, and specify mua_<xxx>_restrictions in main.cf (where
29 # "<xxx>" is "client", "helo", "sender", "relay", or "recipient").
30 # -o smtpd_client_restrictions=
31 # -o smtpd_helo_restrictions=
32 # -o smtpd_sender_restrictions=
33 # -o smtpd_relay_restrictions=
34 # -o smtpd_recipient_restrictions=reject_non_fqdn_recipient,reject_unknown_recipient_domain,permit_sasl_authenticated,reject
35 # -o milter_macro_daemon_name=ORIGINATING
36 # Choose one: enable submissions for loopback clients only, or for any client.
37 #127.0.0.1:submissions inet n - n - - smtpd
```

Рис. 11: Конфигурация master.cf для 587-го порта

```
Timeout : 7200 (sec)
Verify return code: 18 (self-signed certificate)
Extended master secret: no
Max Early Data: 0

---
read R BLOCK
EHLO test
250-server.vabazlov.net
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-AUTH PLAIN
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250-SMTPUTF8
250 CHUNKING
AUTH PLAIN dmFiYXpsb3YAdmFiYXpsb3YAMTIzNDU2
235 2.7.0 Authentication successful
quit
221 2.0.0 Bye
closed
[root@client.vabazlov.net ~]#
```

# Проверка отправки в Evolution



```
Nov 29 08:56:52 server dovecot[23435]: imap-login: Login: user=<vabazlov>, method=PLAIN, rip=192.168.1.30, lip=192.168.1.1, mpid=25431, TLS, session=<Xn1b7bdEPJfAqAE>
Nov 29 08:57:20 server postfix/smtpd[25297]: connect from client.vabazlov.net[192.168.1.30]
Nov 29 08:57:20 server postfix/tlsmgr[25299]: warning: btree:/var/lib/postfix/smtpd_scache is unavailable. unsupported dictionary type: btree
Nov 29 08:57:24 server postfix/smtpd[25297]: CD83D46F93: client=client.vabazlov.net[192.168.1.30], sasl_method=PLAIN, sasl_username=vabazlov
Nov 29 08:57:24 server postfix/cleanup[25502]: CD83D46F93: message-id=<1c96fd152770dda57ccd407543b443f8f05d8266.camel@vabazlov.net>
Nov 29 08:57:24 server postfix/qmgr[25051]: CD83D46F93: from=<vabazlov@vabazlov.net>, size=539, nrcpt=1 (queue active)
Nov 29 08:57:24 server postfix/smtpd[25297]: disconnect from client.vabazlov.net[192.168.1.30] ehlo=2 starttls=1 auth=1 mail=1 rcpt=1 data=1 quit=1 commands=8
Nov 29 08:57:24 server postfix/local[25504]: CD83D46F93: passing <vabazlov@vabazlov.net> to transport=lmtp
Nov 29 08:57:24 server dovecot[23435]: lmtp(25507): Connect from local
Nov 29 08:57:24 server dovecot[23435]: lmtp(vabazlov)<25507><hOL9MnS1KmnjYwAAFrYJw>: msgid=<1c96fd152770dda57ccd407543b443f8f05d8266.camel@vabazlov.net>; saved mail to INBOX
Nov 29 08:57:24 server postfix/lmtp[25505]: CD83D46F93: to=<vabazlov@vabazlov.net>, relay=server.vabazlov.net[private:dovecot-lmtp], delay=0.03, delays=0.01/0.01/0.0, dsn=2.0.0, status=sent (250 2.0.0 <vabazlov@vabazlov.net> hOL9MnS1KmnjYwAAFrYJw Saved)
Nov 29 08:57:24 server postfix/qmgr[25051]: CD83D46F93: removed
Nov 29 08:57:24 server dovecot[23435]: lmtp(25507): Disconnect from local: Logged out (state=READY)
```

Рис. 14: Фрагмент логов доставки письма

## Копирование конфигурационных файлов

```
[root@server.vabazlov.net server]#  
[root@server.vabazlov.net server]# cd /vagrant/provision/server/  
[root@server.vabazlov.net server]# cp -R /etc/dovecot/dovecot.conf /vagrant/provision/server/mail/etc/dovecot/  
cp: overwrite '/vagrant/provision/server/mail/etc/dovecot/dovecot.conf'? y  
[root@server.vabazlov.net server]# cp -R /etc/dovecot/conf.d/10-master.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/  
[root@server.vabazlov.net server]# cp -R /etc/dovecot/conf.d/10-auth.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/  
cp: overwrite '/vagrant/provision/server/mail/etc/dovecot/conf.d/10-auth.conf'? y  
[root@server.vabazlov.net server]# mkdir -p /vagrant/provision/server/mail/etc/postfix  
[root@server.vabazlov.net server]# cp -R /etc/postfix/master.cf /vagrant/provision/server/mail/etc/postfix/  
[root@server.vabazlov.net server]#
```

Рис. 15: Копирование конфигураций

## Обновлённый provisioning-скрипт

```
1  #!/bin/bash
2  echo "Provisioning script $0"
3  echo "Install needed packages"
4  dnf -y install postfix
5  dnf -y install dovecot
6  dnf -y install telnet
7  echo "Copy configuration files"
8  cp -R /vagrant/provision/server/mail/etc/* /etc
9  chown -R root:root /etc/postfix
10 restorecon -vR /etc
11 echo "Configure firewall"
12 firewall-cmd --add-service smtp --permanent
13 firewall-cmd --add-service pop3 --permanent
14 firewall-cmd --add-service pop3s --permanent
15 firewall-cmd --add-service imap --permanent
16 firewall-cmd --add-service imaps --permanent
17 firewall-cmd --add-service smtp-submission --permanent
18 firewall-cmd --reload
19 echo "Start postfix service"
20 systemctl enable postfix
21 systemctl start postfix
22 echo "Configure postfix"
23 postconf -e 'mydomain = vabazlov.net'
24 postconf -e 'myorigin = $mydomain'
25 postconf -e 'inet_protocols = ipv4'
26 postconf -e 'inet_interfaces = all'
27 postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
28 #postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
29 echo "Configure postfix for dovecot"
30 postconf -e 'home_mailbox = Maildir/'
31 echo "Configure postfix for auth"
32 postconf -e 'smtpd_sasl_type = dovecot'
33 postconf -e 'smtpd_sasl_path = private/auth'
34 postconf -e 'smtpd_recipient_restrictions = reject_unknown_recipient_domain, permit_mynetworks,'
35 postconf -e 'mynetworks = 127.0.0.0/8'
36 echo "Configure postfix for SMTP over TLS"
```

Рис. 16: Расширенная версия mail.sh

```
1  #!/bin/bash
2  echo "Provisioning script $0"
3  echo "Install needed packages"
4  dnf -y install postfix
5  dnf -y install s-nail
6  dnf -y install evolution telnet
7
8  echo "Configure postfix"
9  postconf -e 'inet_protocols = ipv4'
10 echo "Start postfix service"
11 systemctl enable postfix
12 systemctl start postfix
```

Рис. 17: Provisioning-скрипт клиента

## Заключение

---

- Настроены LMTP, SMTP-AUTH и TLS в связке Postfix + Dovecot
- Проверены доставка писем и защищённые SMTP-сессии
- Реализована интеграция с системой Vagrant
- Получены навыки настройки защищённого почтового окружения